STATES OF MATTER (BEHAVIOUR OF GASES)

1. According to Charle's law–

(1) $\left(\frac{dV}{dT}\right)_p = k$ (2) $\left(\frac{dV}{dT}\right)_p = -k$

(3) $\left(\frac{dV}{dT}\right)_p = \frac{-k}{T}$ (4) None of these

2. The rate of diffusion of SO_2, CO_2, PCl_3 and SO_3 are in the following order–

(1) $PCl_3 > SO_3 > SO_2 > CO_2$

(2) $CO_2 > SO_2 > PCl_3 > SO_3$

(3) $SO_2 > SO_3 > PCl_3 > CO_2$

(4) $CO_2 > SO_2 > SO_3 > PCl_3$

3. The compressibility factor for H_2 and He is usually

(1) > 1 (2) < 1 (3) = 1 (4) = 0

4. The compressibility factor of an ideal gas

(1) 0 (2) ∞ (3) 1 (4) –1

5. The kinetic energy of molecule is zero at

(1) 0°C (2) 273°C (3) –273°C (4) 116°C

6 The term that correct for the attractive force present in a real gas in the vander waal's equation is

(1) nb (2) $\frac{an^2}{V^2}$ (3) $-\frac{an^2}{V^2}$ (4) – nb

7. If 500 ml of a gas is compressed at constant temperature by increasing its pressure by 10%. Then calculate the final volume

(1) 454.5 mL (2) 354.5 mL

(3) 254.5 mL (4) 154.5 mL

8. Air contains 79% N_2 and 21% O_2 by volume. If the total pressure is 750 mm of Hg. then the partial pressure of O_2 is–

(1) 157.2 mm of Hg

(2) 175.5 mm of Hg

(3) 315.0 mm of Hg

(4) 592.8 mm of Hg

9. Kinetic theory of gases proves

(1) Only Boyle's law (2) Only Charle's law

(3) Only Avogadro's law (4) All of these

10. Which gas can most readly liquified

(1) NH_3 (2) Cl_2 (3) SO_2 (4) CO_2

11. 1 mol CO_2 ocupies 0.4 lit at 27°C and 40 atm calculate the compressibility factor

(1) 0.65 (2) 0.85 (3) 0.15 (D) 1.65

12. Density of a gas is found to be 5.46gm/lit at 27°C and 2 atm pressure. what will be its density at STP

(1) 3 gm/lit (2) 2 gm/lit

(3) 1.5 gm/lit (4) 1 gm/lit